Smart Kitchen Inventory & Prep Management App (Commercial Kitchens)

1. Overview

This project is a commercial kitchen inventory and prep management system designed for cloud kitchens, mid-sized restaurants, and catering operations. The system tracks prep batches, POS orders, portion-controlled recipes, and inventory, ensuring accurate deduction of ingredients and reducing waste.

Target Users: - Cloud kitchens - Mid-sized restaurants - Catering kitchens - Multi-brand / central kitchens

Core Problems Solved: - Inaccurate inventory tracking - Manual stock updates - Overproduction & waste - Lack of visibility across shifts - Managing prep items & batch production

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2. Key Principles

1. **POS-driven inventory deduction**: Orders from POS trigger automatic inventory reduction.
2. **Prep items (sub-recipes)**: Sauces, bases, gravies, dough, marinades, etc. are logged per batch by chefs; raw ingredients are deducted and prep item inventory is increased.
3. **Portion-controlled recipes**: Each menu item has defined ingredient/prep usage per portion; catering tray sizes mapped to ingredient multipliers.
4. **Minimal kitchen disruption**: Prep logging is simple and fast; role-based access ensures proper control.
5. **Waste & variance tracking (Phase 2+)**: Compare expected vs actual usage to identify over-portioning, spoilage, or prep losses.

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3. Data Flow

Step 1: Prep Logging (Chef) - Chef logs batch (e.g., 12 L Tomato Gravy) - Raw ingredients deducted - Prep item inventory increased - Expiry timer starts

Step 2: POS Order Comes In - Menu item sold - System calculates ingredient usage based on portion sizes - Inventory automatically deducted

Step 3: Catering Orders - Tray size selected in POS - Multiplier applied to ingredient usage - Inventory updated automatically

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4. Inventory Structure

- **Raw Ingredients:** Tomato, Onion, Butter, Chicken, etc.
- **Prep Items (Sub-Recipes):** Tomato Gravy, White Sauce, etc.; batch quantity, yield, expiry tracked.
- **Menu Items:** Portion size of each ingredient/prep item defined; catering tray sizes mapped to ingredient multipliers.

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5. User Roles

- **Chef:** Logs prep, views inventory.
- **Kitchen Manager:** Edits recipes, sets portion sizes, approves prep.
- **Admin / Owner:** Sees reports, variance, cost analysis.

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6. MVP Feature List

Must-have: - POS integration (trigger orders) - Recipe-based portion mapping - Prep batch logging - Automatic deduction of prep & raw ingredients - Low-stock & expiry alerts

Optional / Phase 2+: - Waste logging & variance reports - Prep forecasting based on pending orders - Multi-location support - Catering batch suggestions

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7. Major Risks & Phased Solutions

Phase	Risk	Solution
Phase 1	POS Accuracy	Standard portion sizes, automatic deduction, ~85% accuracy accepted
Phase 1	Prep Compliance	Simple prep logging UI, default batch sizes, reminders
Phase 1	Recipe Setup Complexity	Bulk import, cloning templates
Phase 1	Catering Orders	Predefined tray sizes with multipliers
Phase 1	Waste Tracking	Basic manual waste logging
Phase 1	User Adoption	Minimal taps, visible benefits, training

Phase	Risk	Solution
Phase 1	POS Integration	Start with 1–2 POS platforms, fallback CSV import
Phase 2	POS Drift	Optional batch usage logging, daily reconciliation
Phase 2	Waste & Overproduction	Variance dashboards
Phase 2	Custom Catering Orders	Manager-adjustable ingredient quantities
Phase 2	Prep Batch Management	Track batch numbers, expiry, FIFO logic
Phase 2	Forecasting Accuracy	Collect historical POS + prep + waste data
Phase 3	POS Limitations	Optional kitchen-driven logging
Phase 3	Forecasting & Analytics	AI/ML models for prep, ordering, waste prediction
Phase 3	Multi-location Scaling	Centralized dashboards, per-kitchen variance tracking
Phase 3	Adoption & Training	Analytics insights, push reminders for prep

8. Architecture Recommendation

- **Frontend:** Web app, tablet-friendly
- **Backend:** Node.js or FastAPI
- **Database:** PostgreSQL (recipes & portions), DynamoDB (events)
- **Auth:** Role-based access
- **Deployment:** AWS ECS / Lambda
- **Optional:** POS integration

9. Workflow Summary

1. Chef logs prep batch → raw ingredients deducted → prep inventory updated.
2. POS order received → portion sizes applied → inventory deducted automatically.
3. Catering orders → tray size multiplier applied → inventory updated.
4. Manager views inventory, variance, and low-stock alerts.

10. Visual Workflow Diagram

Smart Kitchen Workflow

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Purpose: To provide a timestamped record of project design and architecture for GitHub posting.